Group - 7

The ‘Drim’ Programming Language

Bringing the Classroom Joke to Life via Language Implementation

Course: SWE 4304 (Software Project Lab - I)

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What Are We Making?

The Drim Concept

"Drim" is a **custom-built**, **interpreted** programming language designed to run as a headless application.

The CLI Interpreter

A CLI-based interpreter that reads .drim source files and executes them line-by-line.

Recap on The Architecture

Lexer (The Scanner)

- Reads raw text input and converts it into meaningful "Tokens"
  - transforming "drimming" into a recognizable loop command.

Parser (The Architect)

- Organizes the generated tokens into an Abstract Syntax Tree (AST) to understand logic and math hierarchy.

Interpreter (The Engine)

- Traverses the Abstract Syntax Tree, executing the code in real-time. This is where Drim comes to life!

Features so far

I/O

- Input → “drim()”
- Output → “wake()”

Math Operations

Supports most arithmetic operations including basic (+, -, *, /, %) and Bitwise (and, or, not)

Data Types

Supports INT, FLOAT (double), STRING as data types

Conversion Functions

No need to memorize how to convert between different unit!

Dynamic Typing

Variables seamlessly switch types (e.g., Int ↔ String) on the fly, offering flexibility in data handling.

Branching (If- Else)

Supports branching of block executions using “if-else” statements.

Physics Formula Calculation

Forgot how to calculate data like final velocity? We’ve got that covered! :)

Features to be added

Loops

Loops for multiple iteration of blocks

Scoped Variable

Variables will live in their own block. (Global right now)

Custom Functions

Will support custom function calling with parameters and dynamic return values.

String Interpolation

Formatted output, like

- wake(“Hello {name}”)

Data Structure Library

Custom build built-in data types like

- Stack
- Queue

Technologies Used

Programming Language

C++

Build System

CMake
Cross-platform Make

Version Control

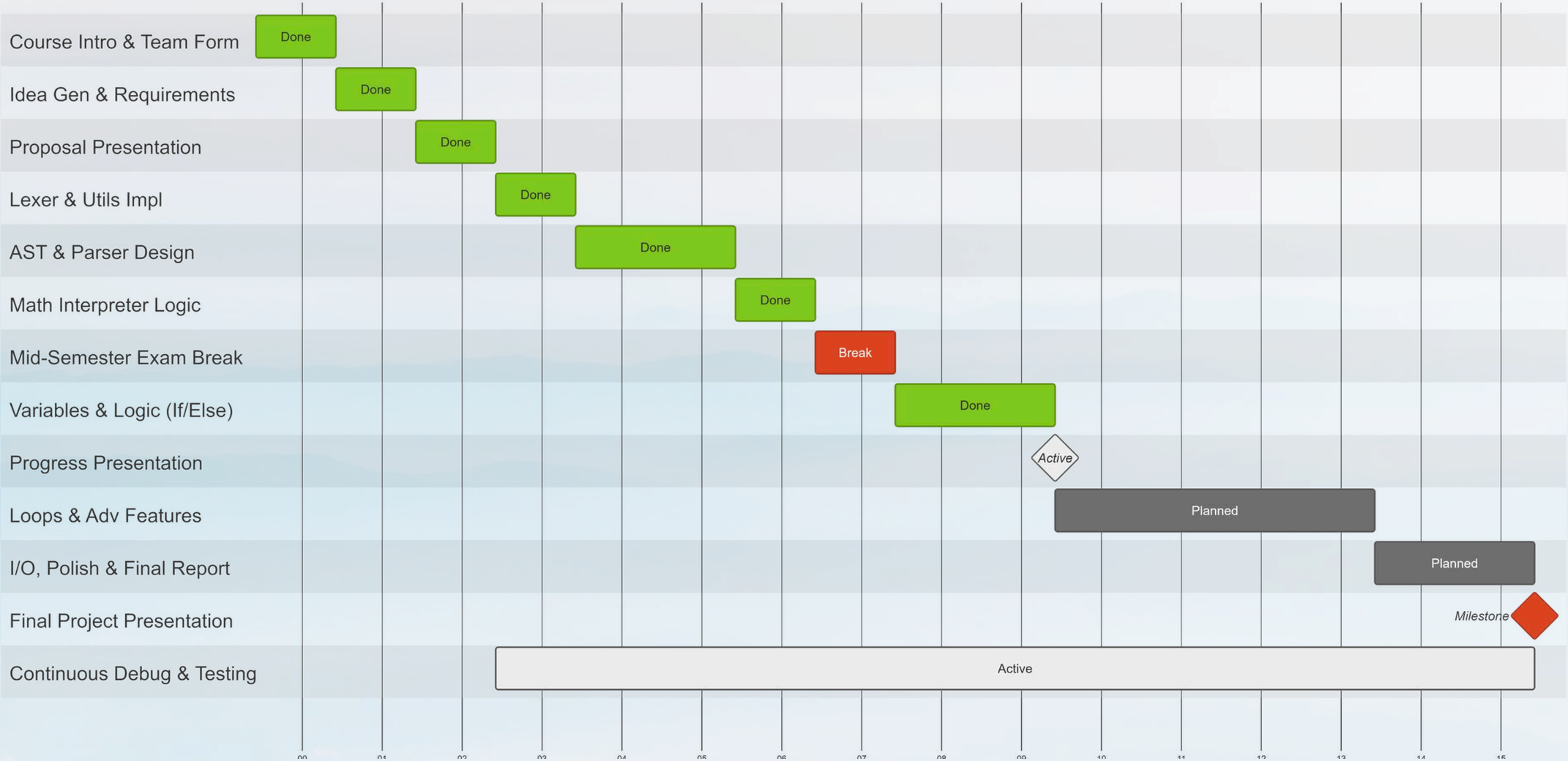
IDE

Visual Studio Code

CL

Diagramming

Current Timeline



Demo

QNA